Career-Technical Credit Transfer (CT)²

Interactive Media Career-Technical Assurance Guide (CTAG)

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Introduction

The following programs/courses, indicated by a Career-Technical Articulation Number (CTAN), are eligible for transfer among Ohio's Public Secondary (CT)2 approved programs/courses and state institutions of higher education. This document presents the following courses in this CTAG:

This CTAG document identifies the learning outcomes that are equivalent or common in introductory technical courses. For students to receive credit under this agreement, the career-technical programs and the state institutions of higher education must document that their course/program content matches the learning outcomes in this CTAG. In accordance with Ohio Revised Code 3333.162, industry standards and certifications provide documentation of student learning.

Accessing Credit

Students that have met the requirements for credit for each CTAG course will appear in the CTAV system that public colleges and universities access to award credit. Institutions must have the student's permission to officially post the CTAG credit to their transcript. Students can grant this permission as part of the end-of-course WebXam. If a student did not grant permission as part of the WebXam, they will need to confirm directly with the colleges and universities they attend that they would like to receive credit for the CTAG course.

Institutional Approval

Secondary: Secondary institutions must have pathway approval from the Ohio Department of Education and Workforce (DEW). Certificate of Affirmation assurances are now incorporated into the CTE-26 application process.

Colleges/Universities: Postsecondary institutions must submit courses through CEMS for review and approval by a faculty panel.

CTIM001 2-D Animation	**Credits: 3 Semester Hours**

Course Description

2-D Animation focuses on the creation and distribution of interactive, computer-based animations. Students create tweened animations as well as interactive animations. Students gain basic knowledge of common scripting languages.

Learning Outcomes

*All learning outcomes that are considered essential are marked with an asterisk.

1. *Identify the interface and tools of industry standard 2-D Animation authoring software.
2. *Create 2-D animations applying the tools and interface of industry standard authoring software.
3. *Create interactive 2D animations applying the tools and interface of industry standard authoring software.
4. *Perform editing techniques within 2-D animations.
5. *Integrate 2-D animations with other applications.
6. *Generate animations in the appropriate output format for intended use.
7. *Incorporate media elements such as sound, video, vector, and raster graphics in a 2-D animation.

Requirements for Credit

- Successfully complete DEW secondary course **Animation (145115)** and earn a qualifying score on the corresponding End-of-Course exam.
- Within 3 years of completing the approved secondary program, matriculate to an institution of higher education with an approved or comparable program.

Learning Outcome Alignment

Post-Secondary Learning Outcomes The student will be able to:	Aligned Secondary Competencies in DEW Career Technical Content Standards
1. *Identify the interface and tools of industry standard 2-D Animation authoring software.	7.1.1. Identify the types and uses of interactive media environments (e.g., web-based, kiosks, games, mobile devices, video, print). 7.1.2. Describe the components of interactive media.
2. *Create 2-D animations applying the tools and interface of industry standard authoring software.	7.6.2. Create and import 2D assets and environments.
3. *Create interactive 2D animations applying the tools and interface of industry standard authoring software.	7.2.1. Develop navigational structures, wireframes, and flowcharts for multimedia applications. 7.2.2. Construct and place navigational units. 7.2.3. Build in interactive elements. 7.2.4. Determine uses and needs for site maps, multimedia scripts, storyboards, and flowcharts. 7.2.6. Place buttons and navigational graphics. 7.6.4. Create special effects and virtual navigation.
4. *Perform editing techniques within 2-D animations.	7.6.2. Create and import 2D assets and environments. 7.6.3. Create key frames and apply tweens and paths. 7.6.4. Create special effects and virtual navigation.
5. *Integrate 2-D animations with other applications.	7.3.1. Select the media elements to be used (e.g., sound, video, graphics, text, animation). 7.3.2. Generate text for multi-image presentations (e.g., title graphics, charts, graphs). 7.3.3. Incorporate graphics (e.g., digital, hand drawn, photographic).
6. *Generate animations in the appropriate output format for intended use.	7.6.6. Render and export animations.
7. *Incorporate media elements such as sound, video, vector, and raster graphics in a 2-D animation.	7.3.3. Incorporate graphics (e.g., digital, hand drawn, photographic). 7.3.4. Incorporate computer animation. 7.3.6. Incorporate video footage.

	7.3.8. Record and/or acquire soundtrack (e.g., narrative, voiceover, sound effects, music). 7.3.9. Integrate sound with visuals. 7.3.10. Produce, test, debug, and archive final product.

CTIM004 Introduction to Web Development	**Credits: 3 Semester Hours**

Course Description

This is an introductory course in Internet technologies and creating Web sites using markup, styling, and scripting languages such as HTML and CSS. Students understand the general nature, function, and structure of the Internet and World Wide Web and develop a simple, static, Web site using a text editor.

Learning Outcomes

This course aligns with TAG course OIT003 Introduction to Web Development.

*All learning outcomes that are considered essential are marked with an asterisk.

1. *Understand foundations, resources and applications of the Internet and the world-wide web.
2. *Build websites using current HTML standards to establish document structure and content of webpages.
3. *Apply CSS elements to describe the presentation of webpages.
4. *Effectively incorporate multimedia files for websites.
5. *Develop basic programming skills using interactive client-side programming language to enhance webpages.
6. *Apply other appropriate industry tools, technologies, and web standards from standards bodies for web development.

Requirements for Credit

- Successfully complete DEW secondary course **Web Design (145010)** and earn a qualifying score on the corresponding End-of-Course exam.
- Within 3 years of completing the approved secondary program, matriculate to an institution of higher education with an approved or comparable program.
- Please note that this course is NOT a pre-requisite for Graphical Web Design. Students need not fulfill the requirements for this class to receive credit for the Graphical Web design class if they meet the outcomes of Graphical Website Design.

Learning Outcome Alignment

Post-Secondary Learning Outcomes The student will be able to:	Aligned Secondary Competencies in DEW Career Technical Content Standards
1. *Understand foundations, resources and applications of the Internet and the world-wide web.	6.1.1. Describe the basic principles of Hypertext Markup Language (HTML) and its functional relationship with web browsers. 6.1.2. Plan a webpage considering subject, devices, audience, layout, color, links, graphics, and Americans with Disabilities Act (ADA) requirements.
2. *Build websites using current HTML standards to establish document structure and content of webpages.	6.1.5. Create and format ordered and unordered lists on a webpage using HTML list formatting tags. 6.1.6. Create and format a table in a webpage using HTML table formatting tags and attributes. 6.1.7. Integrate styles (e.g., inline or external Cascading Style Sheets [CSS]). 6.2.1. Create absolute links and relative links. 6.2.2. Write a Hypertext Markup Language (HTML) anchor that links to another section of the same web page. 6.2.3. Create hyperlinks that send e-mail messages and download files. 6.2.4. Insert image and wrap text around the image using Cascading Style Sheets (CSS). 6.2.5. Resize a graphic image in a web page using CSS. 6.2.6. Insert media files (e.g., audio, video,) into a web page using HTML tags. 6.2.7. Build a hover or mouseover effect to change the style of a link. 6.4.1. Design a data entry form from specifications that will accept a variety of user inputs (e.g., radio buttons, text entry fields, check boxes, drop-down menus). 6.4.2. Write the Hypertext Markup Language (HTML) code to add a form to a webpage. 6.4.3. Write the HTML code to add text entry fields, radio buttons, check boxes, drop-down menus, and other user inputs to a form. 6.4.4. Explain the concept of a form action. 6.4.5. Write the HTML code to add a working button (e.g., submit, reset) to a form.

	6.4.6. Format a completed form using HTML and Cascading Style Sheets (CSS) (e.g., fieldset, tabindex). 6.4.7. Code scripting to interact with data sources (e.g., database, web services).
3. *Apply CSS elements to describe the presentation of webpages.	6.3.1. Select and apply scripting languages used in web development. 6.5.3. Utilize standard web programming languages (e.g., markup, scripting languages) in website development.
4. *Effectively incorporate multimedia files for websites.	6.1.5. Create and format ordered and unordered lists on a webpage using HTML list formatting tags. 6.1.6. Create and format a table in a webpage using HTML table formatting tags and attributes. 6.1.7. Integrate styles (e.g., inline or external Cascading Style Sheets [CSS]). 6.2.1. Create absolute links and relative links. 6.2.2. Write a Hypertext Markup Language (HTML) anchor that links to another section of the same web page. 6.2.3. Create hyperlinks that send e-mail messages and download files. 6.2.4. Insert image and wrap text around the image using Cascading Style Sheets (CSS). 6.2.5. Resize a graphic image in a web page using CSS. 6.2.6. Insert media files (e.g., audio, video,) into a web page using HTML tags. 6.2.7. Build a hover or mouseover effect to change the style of a link. 6.4.1. Design a data entry form from specifications that will accept a variety of user inputs (e.g., radio buttons, text entry fields, check boxes, drop-down menus). 6.4.2. Write the Hypertext Markup Language (HTML) code to add a form to a webpage. 6.4.3. Write the HTML code to add text entry fields, radio buttons, check boxes, drop-down menus, and other user inputs to a form. 6.4.4. Explain the concept of a form action. 6.4.5. Write the HTML code to add a working button (e.g., submit, reset) to a form. 6.4.6. Format a completed form using HTML and Cascading Style Sheets (CSS) (e.g., fieldset, tabindex). 6.4.7. Code scripting to interact with data sources (e.g., database, web services).

5. *Develop basic programming skills using interactive client-side programming language to enhance webpages.	6.3.1. Select and apply scripting languages used in web development. 6.5.3. Use standard web programming languages (e.g., markup, scripting languages) in website development.
6. *Apply other appropriate industry tools, technologies, and web standards from standards bodies for web development.	6.1.2. Plan a webpage considering subject, devices, audience, layout, color, links, graphics, and Americans with Disabilities Act (ADA) requirements. 6.5.1. Implement web programming standards and protocols (e.g., World Wide Web Consortium [W3C], Hypertext Markup Language [HTML] 5). 6.5.2. Plan a website's structure for navigation and usability. 6.5.3. Use standard web programming languages (e.g., markup, scripting languages) in website development. 6.5.4. Install and configure a content management system (CMS). 6.5.5. Select an integrated development environment (IDE).

CTIM005 Graphical Website Design — Credits: 3 Semester Hours

Course Description

Graphical Website Design concentrates on the development and Management of websites using a WYSIWYG website management tool (e.g., Squarespace, Wix, Wordpress, Webflow, Dreamweaver). Students create a multimedia website from the project planning stage through usability testing.

Learning Outcomes

*All learning outcomes that are considered essential are marked with an asterisk.

1. *Understand customer needs, information requirements and project scope for Web site development.
2. *Create, customize, and maintain a Web site using industry standard web development and management software.
3. *Create and customize a page layout using industry standard web development and management software.
4. *Add media elements to a Web site.
5. *Create a website usability testing process.

Requirements for Credit

- Successfully complete DEW secondary course **Interactive Application Development (145125)** and earn a qualifying score on the corresponding End-of-Course exam.
- Within 3 years of completing the approved secondary program, matriculate to an institution of higher education with an approved or comparable program.

Learning Outcome Alignment

Post-Secondary Learning Outcomes The student will be able to:	Aligned Secondary Competencies in DEW Career Technical Content Standards
1. *Understand customer needs, information requirements and project scope for Web site development.	7.1.1. Identify the types and uses of interactive media environments (e.g., web-based, kiosks, games, mobile devices, video, print). 7.1.8. Analyze the social and cultural implications of interactive media. 7.1.9. Identify major applications for interactive media (e.g., sales and marketing, interactive advertising, education, online learning, corporate training, corporate communications, news, entertainment). 7.1.10. Identify specific uses for interactive media in potential markets. 7.2.5. Make preliminary sketches showing placement of images and text on screen.
2. *Create, customize, and maintain a Web site using industry standard web development and management software.	6.1.3. Format the content of a web page using HTML formatting tags (e.g., hyperlink, e-mail, table formatting, graphic attributes). 6.1.5. Create and format ordered and unordered lists on a webpage using HTML list formatting tags. 6.1.6. Create and format a table in a webpage using HTML table formatting tags and attributes. 6.1.7. Integrate styles (e.g., inline or external Cascading Style Sheets [CSS]). 6.2.1. Create absolute links and relative links. 6.2.3. Create hyperlinks that send e-mail messages and download files. 6.2.4. Insert image and wrap text around the image using Cascading Style Sheets (CSS). 6.2.5. Resize a graphic image in a webpage using CSS. 6.2.6. Insert media files (e.g., audio, video) into a web page using HTML tags. 6.2.7. Build a hover or mouseover effect to change the style of a link. 6.4.1. Design a data entry form from specifications that will accept a variety of user inputs, (e.g., radio buttons, text entry fields, check boxes, drop- down menus). 6.4.4. Explain the concept of a form action. 6.4.6. Format a completed form using HTML and Cascading Style Sheets (CSS) (e.g., fieldset, tabindex). 6.5.4. Install and configure a content management system (CMS).

3. *Create and customize a page layout using industry standard web development and management software.	2.7.4. Identify how the use of different browsers and devices effects the function of a webpage (e.g., Americans with Disabilities Act [ADA], text-to-speech, screen reader, mobile vs. desktop). 6.5.1. Implement web programming standards and protocols (e.g., World Wide Web Consortium [W3C], Hypertext Markup Language [HTML]5). 6.5.2. Plan a website's structure for navigation and usability. 6.5.3. Utilize standard web programming languages (e.g., markup, scripting languages) in website development. 6.5.5. Select an integrated development environment (IDE). 6.5.6. Create and edit a webpage template. 6.5.7. Create and attach Cascading Style Sheets (CSS). 6.5.8. Format website layout (e.g., targeted platforms, text formatting, background color, text, tables, lists, iframes). 6.5.9. Incorporate audio and video, forms, and links on a website. 6.5.10. Develop and execute usability tests on a completed website, checking for information accessibility, ease of use, and navigation. 6.5.12. Publish the completed website to a web server.
4. *Add media elements to a Web site.	6.5.9. Incorporate audio and video, forms, and links on a website. 7.2.9. Provide a sample layout to stakeholders for review. 7.2.10. Select and create visual design elements appropriate for the intended audience and use.
5. *Create a website usability testing process.	6.5.2. Plan a website's structure for navigation and usability. 6.5.10. Develop and execute usability tests on a completed website, checking for information accessibility, ease of use, and navigation.

CTIM006 Digital Video Production	**Credits: 3 Semester Hours**

Course Description

Digital Video Production focuses on the development of video from the preproduction process through the production and post-production phases. Students plan, shoot, edit, and distribute videos as part of a production team. Topics include preparing a script, developing a shot list, videography, editing footage, adding soundtracks, and exporting and rendering video for various uses in various formats. This course may use several editing platforms and equipment.

Learning Outcomes

*All learning outcomes that are considered essential are marked with an asterisk.

1. *Use video pre-production process.
2. *Use video cameras, camcorders, and other equipment.
3. *Identify and select the most appropriate video format.
4. *Edit digital video and include various supporting media elements.
5. *Import and export digital video.

Requirements for Credit

- Successfully complete DEW secondary course **Video and Sound (145110)** and earn a qualifying score on the corresponding End-of-Course exam.
- Within 3 years of completing the approved secondary program, matriculate to an institution of higher education with an approved or comparable program.

Learning Outcome Alignment

Post-Secondary Learning Outcomes The student will be able to:	Aligned Secondary Competencies in DEW Career Technical Content Standards
1. *Use video pre-production process.	7.7.1. Identify equipment and other production needs (e.g. drone, stop action, Digital Single Lens Reflex (DSLR), mirrorless, compact, and 360 cameras). 7.7.2. Analyze the script and storyboard to develop a production schedule. 7.7.5. Select a linear or nonlinear editing system and edit the video.
2. *Use video cameras, camcorders, and other equipment.	7.7.4. Select a video recording format and shoot the video.
3. *Identify and select the most appropriate video format.	7.7.4. Select a video recording format and shoot the video.
4. *Edit digital video and include various supporting media elements.	7.3.7. Edit video footage. 7.7.6. Add transitions (e.g., dissolves, wipes, cuts), titles, special effects, and digital effects. 7.7.7. Add a soundtrack, narration, and/or voiceover. 7.7.8. Export video to the desired medium.
5. *Import and export digital video.	7.3.6. Incorporate video footage. 7.7.8. Export video to the desired medium.

CTIM007 3-D Modeling and Animation	Credits: 3 Semester Hours

Course Description

This course covers the basics of 3-D modeling and animation techniques, including lighting/shadow, textures, and cameras. Students learn to identify and use an industry standard 3-D development environment to create characters and graphics.

Learning Outcomes

*All learning outcomes that are considered essential are marked with an asterisk.

1. *Identify the interface and tools of industry standard 3-D Animation development environments.
2. *Create 3D animations applying the tools and principles of 3-D modeling.
3. *Create, texture, and render 3-D models using industry standard tools and software.
4. *Apply 3-D animation and image generation techniques.

Requirements for Credit

- Successfully complete DEW secondary course **Hard Surface Modeling (145140)** and earn a qualifying score on the corresponding End-of-Course exam.
- Within 3 years of completing the approved secondary program, matriculate to an institution of higher education with an approved or comparable program.

Learning Outcome Alignment

Post-Secondary Learning Outcomes The student will be able to:	Aligned Secondary Competencies in DEW Career Technical Content Standards
1. *Identify the interface and tools of industry standard 3-D Animation development environments.	7.6.3. Create key frames and apply tweens and paths. 7.6.4. Create special effects and virtual navigation. 7.6.5. Create and import 3D assets and environments. 7.6.6. Render and export animations.
2. *Create 3D animations applying the tools and principles of 3-D modeling.	7.6.5. Create and import 3D assets and environments. 7.6.7. Create and import virtual assets and environments.
3. *Create, texture, and render 3-D models using industry standard tools and software.	7.6.5. Create and import 3D assets and environments. 7.6.6. Render and export animations. 7.6.8. Create and render materials in a 3D environment.
4. *Apply 3-D animation and image generation techniques.	7.6.3. Create key frames and apply tweens and paths. 7.6.6. Render and export animations. 7.6.9. Create 3D shapes through box modeling.

CTIM008 Digital Page Layout and Design	Credits: 3 Semester Hours

Course Description

Digital Page Layout and Design focuses on the creation of print and digital publications using industry-standard page-layout software. Students learn to plan, create, and distribute digital publications.

Learning Outcomes

*All learning outcomes that are considered essential are marked with an asterisk.

1. *Research and plan page-layout projects.
2. *Choose appropriate media elements to meet project output needs.
3. *Demonstrate an understanding of color models.
4. *Apply graphic design principles and elements.
5. *Integrate knowledge of typography.
6. Generate documents in the appropriate file format for intended use.

Requirements for Credit

- Successfully complete DEW secondary course **Design Techniques (145095)** and earn a qualifying score on the corresponding End-of-Course exam.
- Within 3 years of completing the approved secondary program, matriculate to an institution of higher education with an approved or comparable program.

Learning Outcome Alignment

Post-Secondary Learning Outcomes The student will be able to:	Aligned Secondary Competencies in DEW Career Technical Content Standards
1. *Research and plan page-layout projects.	2.9.1. Identify the scope and purpose of branding. 2.9.2. Determine the scope and purpose of the project. 2.9.3. Determine the target audience, client needs, expected outcomes, objectives, and budget. 2.9.4. Develop a conceptual model and design brief for the project. 2.9.5. Develop a timeline, communication plan, task breakdown, costs (e.g., equipment, labor), deliverables, and responsibilities for completion. 2.9.6. Develop and present a comprehensive proposal to stakeholders. 7.2.5. Make preliminary sketches showing placement of images and text on screen. 7.2.9. Provide a sample layout to stakeholders for review. 7.2.10. Select and create visual design elements appropriate for the intended audience and use.
2. *Choose appropriate media elements to meet project output needs.	7.2.10. Select and create visual design elements appropriate for the intended audience and use. 7.4.2. Select color, shape, size, and texture of objects. 7.4.3. Create or acquire graphics. 7.4.4. Manipulate and layer objects. 7.4.5. Differentiate between vector and raster images. 7.4.6. Select graphic software applications based on budget, technical capabilities, and hardware specifications to meet intended project outcome. 7.4.7. Optimize and export graphics files for intended use. 7.4.9. Compress and decompress graphic files.
3. *Demonstrate an understanding of color models.	7.2.7. Select colors based on color theory and psychology. 7.4.1. Select and manipulate color profiles (e.g., Red Green Blue [RGB], Cyan Magenta Yellow Key [CMYK], Pantone) for appropriate uses. 7.4.10. Describe and select color profiles (e.g., Red Green Blue [RGB], Cyan Magenta Yellow Key [CMYK], Pantone).

4. *Apply graphic design principles and elements.	7.3.2. Generate text for multi-image presentations (e.g., title graphics, charts, graphs). 7.3.3. Incorporate graphics (e.g., digital, hand drawn, photographic).
5. *Integrate knowledge of typography.	7.5.1. Identify typographic measurements (e.g., picas, points, pixels, ems). 7.5.2. Mix families of type within a project. 7.5.3. Select appropriate kerning, leading, tracking, and other related formatting. 7.5.4. Identify appropriate typefaces (e.g., serif, sans serif, Web Safe, screen, print). 7.5.5. Prepare a type style guide.
6. Generate documents in the appropriate file format for intended use.	7.4.7. Optimize and export graphics files for intended use. 7.4.9. Compress and decompress graphic files.

Additional Requirements and Credit Conditions
1. The receiving institution must have a comparable program, major, or course that has been approved through submission to the Ohio Department of Higher Education (CT) ^{2} approval process for the CTAN listed in this document. 2. Credits apply to courses in the specified technical area at Ohio's public institutions of higher education, if the institution offers courses in the specific technical area. In the absence of an equivalent course, and when the institution offers the technical program, the receiving institution is encouraged to grant and apply an equivalent credit value of the Career-Technical Articulation Number (CTAN) toward the technical requirements of the specific degree/certificate program. 3. A career-technical student seeking credit under the terms of this CTAG must apply and be accepted to the college within three years of completing a career-technical course, unless otherwise noted. 4. A career-technical student who meets all eligibility criteria will receive the credit hour value for the comparable course as offered at the receiving state institution of higher education. 5. The admission requirements of individual institutions and/or programs are unaffected by the implementation of (CT)^2 outcomes. 6. The transfer of credit, through this CTAG, will not exempt a student from the residency requirements at the receiving institution.

Interactive Media Faculty Panels

Original Panel Participants, 2015		
Justin McCrea	Stark State College	Lead Panel Expert
Tom Dulaney	Eastern Gateway Community College	Lead Panel Expert
Dave Hubble	University of Cincinnati	Panel Expert
Brittney Miller	Central Ohio Technical College	Panel Expert
Cindy Berelsman	Wright State University	Panel Expert
Cyndi Brill	Ohio Department of Education	Program Specialist
Aaron Stewart	Ohio Department of Education	Program Specialist
Dr. Robert Haas	Ohio Department of Higher Education	SCTAI Special Coach
Jamilah Tucker	Ohio Department of Higher Education	Director of Career-Technical Transfer Initiatives
Anne Skuce	Ohio Department of Higher Education	Senior Associate Director, SCTAI
Misty McKee	Ohio Department of Higher Education	Assistant Director, SCTAI
Jessi Spencer	Ohio Department of Higher Education	Administrative Coordinator of SCTAI

Review Panel Participants, 2025		
Andy Curran	University of Cincinnati	Lead Panel Expert
Dovel Myers	Shawnee State University	Lead Panel Expert (CTIM004 only)
Michelle Lash	Stark State	Panel Expert
Amanda Romero	Sinclair Community College	Panel Expert
Scott Dawson	Clark State Community College	Panel Expert
Jon Lundquist	Columbus State Community College	Panel Expert
Kristopher Vigneron	Ohio Department of Education and Workforce	Program Specialist
Brittany Kim	Ohio Department of Higher Education	Associate Director, SCTAI